

26/03/2025

MODULE - III

INDUCTION AND RECURRENCES

Mathematical Induction

$$1 + 2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

→ To prove that $p(n)$ is true for all positive integers n , we complete two steps:

① BASIS STEP: We verify that $p(1)$ is true.

② INDUCTIVE STEP: Here, we assume that the result is proved for $n=k$, i.e., $p(k)$ is true.

Then, we have to verify $p(k+1)$ is true.

Q1 Use mathematical induction to prove the following:

① $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

Let $p(n): 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

Basis Step

$p(1), \text{ LHS} = 1$ $\frac{1 \times 2 \times 1}{2} = \frac{(1+1)(1)}{2} = 1$

$\text{RHS} = \frac{1(1+1)}{2} = 1$

$\therefore \text{LHS} = \text{RHS}$

$\therefore p(1)$ is true

$$(1+2+\dots+n) + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{n(n+1) + 2(n+1)}{2} = \frac{(n+1)(n+2)}{2}$$

Inductive step

we assume $p(k)$ is true, i.e.,

$$p(k),$$

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

Now, we have to prove $p(k+1)$ is true.

$$\begin{aligned} p(k+1): 1 + 2 + 3 + 4 + \dots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \end{aligned}$$

$$= \frac{(k+1)(k+2)}{2}$$

which is true.

$\therefore$ by induction, $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

$$\textcircled{2} \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

let $p(n): 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Basis step

$$p(1),$$

$$\text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1(1+1)(2+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore p(1)$ is true

Inductive step

we assume $p(k)$ is true, i.e.,

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

Now, we have to prove $p(k+1)$ is true.

$$\begin{aligned}
 p(k+1) &: 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 \\
 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\
 &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\
 &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\
 &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\
 &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\
 &= \frac{(k+1)(2k^2 + 4k + 3k + 6)}{6} \\
 &= \frac{(k+1)[2k(k+2) + 3(k+2)]}{6} \\
 &= \frac{(k+1)(k+2)(2k+3)}{6}, \text{ which is true.}
 \end{aligned}$$

$\therefore$ by induction, $1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$

$$\textcircled{3} \quad 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

let $p(n): 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$

Basis step

$$p(1), \text{ LHS} = 1^3 = 1$$

$$\text{RHS} = \left(\frac{1(1+1)}{2} \right)^2 = 1^2 = 1$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore p(1)$ is true

Inductive step

we assume $p(k)$ is true, i.e.,

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2$$

$$\left(\frac{(k+1)(k+2)}{2} \right)^2$$

we have to prove $p(k+1)$ is also true,

$$p(k+1): 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3$$

$$= \frac{k^2(k+1)^2 + 4(k+1)^3}{4}$$

$$= \frac{(k+1)^2 [k^2 + 4k + 4]}{4}$$

$$= \frac{(k+1)^2 (k+2)^2}{4}$$

$$= \left[\frac{(k+1)(k+2)}{2} \right]^2, \text{ which is true}$$

$\therefore$ by induction, $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$

$$(4) \quad 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n \cdot (n+1) = \frac{n(n+1)(2n+2)}{3}$$

$$\text{let } p(n): 1 \cdot 2 + 2 \cdot 3 + \dots + n \cdot (n+1) = \frac{n(n+1)(2n+2)}{3} \quad (8)$$

Basis step

$$p(1), \text{ LHS} = 1 \cdot 2 = 2$$

$$\text{RHS} = \frac{1(2)(3)}{3} = \underline{\underline{2}}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore p(1)$ is true

Inductive step

we assume $p(k)$ is true, i.e.,

$$1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

we have to prove that $p(k+1)$ is also true,

$$p(k+1) : 1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3}$$

$$= \frac{(k+1)(k+2)(k+3)}{3}, \text{ which is true.}$$

$$\therefore \text{by induction, } 1 \cdot 2 + 2 \cdot 3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$$\textcircled{5} \quad 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

$$\text{let } p(n) : 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

Basis step

$p(1)$,

$$\text{LHS} = 1 \cdot 2 \cdot 3 = 6$$

$$\text{RHS} = \frac{1(2)(3)(4)}{4} = 6$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore p(1)$ is true

Inductive step

We assume $p(k)$ is true, that is,

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + k(k+1)(k+2) = \frac{k(k+1)(k+2)(k+3)}{4}$$

Then, $p(k+1)$:

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + k(k+1)(k+2) + (k+1)(k+2)(k+3)$$
$$= \frac{k(k+1)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3)$$
$$= \frac{(k+1)(k+2)(k+3)[k+4]}{4}, \text{ which is true}$$

$\therefore$ by induction, $1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$

⑥ $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$, $\forall$ non-negative integers n

Let $p(n): 1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

Basis step

$$p(0) = 2^0 = 1 \text{ LHS} = 1$$

$$\text{RHS} = 2^{0+1} - 1 = 2 - 1 = 1$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore p(0) \text{ is true}$$

Inductive step

We assume $p(k)$ is true, i.e.,

$$1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1$$

$$\begin{aligned} \text{Then, } p(k+1) &: 1 + 2^1 + 2^2 + \dots + 2^k + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1} \\ &= 2(2^{k+1}) - 1 \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1, \text{ which is true.} \end{aligned}$$

$\therefore$ by induction, $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

④ prove that 21 divides $4^{n+1} + 5^{2n-1}$ whenever n is a positive integer.

let $p(n)$: 21 divides $4^{n+1} + 5^{2n-1}$

Basis

$p(1)$:

$$4^{1+1} + 5^{2 \cdot 1 - 1} = 16 + 5 = 21 \text{ which is divisible by 21}$$

i.e., 21 divides 21, which is true.

So, $p(1)$ is true.

Inductive

Assume $p(k)$ is true, i.e.,

$$21 \text{ divides } 4^{k+1} + 5^{2k-1}$$

$$\text{i.e., } 4^{k+1} + 5^{2k-1} = 21 \times m, \text{ where } m \text{ is a constant.}$$

$$\text{Then, } p(k+1): 21 \text{ divides } 4^{k+2} + 5^{2k+2-1}$$

$$\begin{aligned} \text{Consider, } 4^{k+2} + 5^{2k+2-1} &= 4 \cdot 4^{k+1} + 25 \cdot 5^{2k-1} \\ &= 4 \cdot 4^{k+1} + (21 + 4) \cdot 5^{2k-1} \\ &= 4 \cdot 4^{k+1} + 21 \cdot 5^{2k-1} + 4 \cdot 5^{2k-1} \\ &= 4(4^{k+1} + 5^{2k-1}) + 21 \cdot 5^{2k-1} \end{aligned}$$

Substituting ①

$$= 4(21 \times m) + 21 \cdot 5^{2k-1}$$

$$= 21(4m + 5^{2k-1})$$

therefore, $p(k+1)$ is true

$\therefore$ by induction, 21 divides $4^{n+1} + 5^{2n+1}$

Hence, proved

⑧ Use induction to PT $7^{n+2} + 8^{2n+1}$ is divisible by 57 $\forall$ non-negative integer.

⑨ PT $n^3 + 2n$ is divisible by 3 $\forall$ non-negative integer.

⑩ PT $n^3 - n$ is divisible by 3 $\forall$ positive integer.

⑪ PT if n is a positive integer, 133 divides $11^{n+1} + 12^{2n+1}$.

Answers.

⑧ $p(n): 7^{n+2} + 8^{2n+1}$ is divisible by 57.

Basis.

$$p(0): 7^2 + 8 \text{ is divisible by } 57$$
$$= 49 + 8$$
$$= 57, \text{ which is divisible by } 57.$$

$\therefore p(0)$ is true

Inductive

Assume $p(k)$ is true, i.e.,

$7^{k+2} + 8^{2k+1}$ is divisible by 57.

$$\Rightarrow 7^{k+2} + 8^{2k+1} = 57 \times m, \quad m \text{ is a constant}$$

①

For $P(k+1)$: $7^{k+3} + 8^{2k+3}$ is divisible by 57. (1)

$$\Rightarrow 7 \cdot 7^{k+2} + 8^2 \cdot 8^{2k+1} \text{ is divisible by } 57.$$

$$\Rightarrow 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1}$$

$$\Rightarrow 7 \cdot 7^{k+2} + (7+57) 8^{2k+1}$$

$$\Rightarrow 7 \cdot 7^{k+2} + 7 \cdot 8^{2k+1} + 57 \cdot 8^{2k+1}$$

$$\Rightarrow 7(7^{k+2} + 8^{2k+1}) + 57 \cdot 8^{2k+1}$$

Substituting (1) $\Rightarrow 7(57 \times m) + 57 \cdot 8^{2k+1}$

$$= 57(7m + 8^{2k+1}), \text{ which is divisible by } 57$$

$\therefore$ by induction, $7^{n+2} + 8^{2n+1}$ is divisible by 57

Hence, proved

(9) let $P(n)$: $n^3 + 2n$ is divisible by 3

Basis

$$P(0) \div 0+0 \text{ is divisible by } 3$$

$$\Rightarrow 0 \text{ is divisible by } 3, \text{ which is true.}$$

$\therefore P(0)$ is true

Inductive

Assume $P(k)$ is true, i.e., $k^3 + 2k$ is divisible by 3.

Then, $k^3 + 2k = 3 \times m$, where m is a constant.

Then $P(k+1)$: $(k+1)^3 + 2(k+1)$ is divisible by 3.

$$(k+1)^3 + 2(k+1) = k^3 + 1 + 3k(k+1) + 2k + 2$$

$$= k^3 + 2k + 3k(k+1) + 3$$

$$= 3 \times m + 3k(k+1) + 3$$

Substituting (1),

$$= 3(m + k(k+1) + 1), \text{ is divisible by } 3.$$

$\therefore$ by induction, $n^3 + 2n$ is divisible by 3.

Hence, proved

⑩ let $p(n): n^3 - n$ is divisible by 3. $\therefore (1^3 - 1) = 0$

Basis

$p(1): 1 - 1 = 0$ is divisible by 3,

$\therefore p(1)$ is true

Inductive

Assume $p(k)$ is true, $k^3 - k$ is divisible by 3.

Then, $k^3 - k = 3 \times m$, m is a constant.

①

Then $p(k+1): (k+1)^3 - (k+1)$, is divisible by 3.

$$= (k+1)^3 - (k+1)$$

$$= k^3 + 1 + 3k(k+1) - (k+1)$$

$$= k^3 + 3k^2 + 3k + 1 - k - 1$$

$$= k^3 - k + 3k(k+1)$$

Substituting, $= 3 \times m + 3k(k+1)$

①

$$= 3(m + k + 1), \text{ which is divisible by 3.}$$

Therefore, $p(k+1)$ is true

$\therefore$ by induction, $n^3 - n$ is divisible by 3.

Hence, proved

⑪ let $p(n): 133$ divides $11^{n+1} + 12^{2n-1}$

Basis

$$p(1): (133 \text{ divides } 11^2 + 12^2 = 121 + 144 = 265)$$

133 divides 133, which is true.

So, $p(1)$ is true

Inductive

Assume $p(k)$ is true, i.e., $11^{k+1} + 12^{2k+1}$ is divisible by 133.

Then, $11^{k+1} + 12^{2k+1} = 133 \times m$, m is constant. ①

$$\begin{aligned} \text{Then } p(k+1): 133 \text{ divides } & 11^{k+2} + 12^{2k+2} \\ &= 11 \cdot 11^{k+1} + 12^2 \cdot 12^{2k+1} \\ &= 11 \cdot 11^{k+1} + 144 \cdot 12^{2k+1} \\ &= 11 \cdot 11^{k+1} + (133 + 11) \cdot 12^{2k+1} \\ &= 11 \cdot 11^{k+1} + 133 \cdot 12^{2k+1} + 11 \cdot 12^{2k+1} \\ &= 11(11^{k+1} + 12^{2k+1}) + 133 \cdot 12^{2k+1} \\ &= 11(133m) + 133 \cdot 12^{2k+1} \end{aligned}$$

$$= 133(11m + 12^{2k+1}),$$

which is divisible by 133.

$\therefore p(k+1)$ is true

$\therefore$ by induction, 133 divides $11^{n+1} + 12^{2n+1}$

Hence, proved

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Q. Prove by induction that $a + ax + ax^2 + \dots + ax^n = \frac{ax^{n+1} - a}{x - 1}$, $x \neq 1$

let $P(n): a + ax + ax^2 + \dots + ax^n = \frac{ax^{n+1} - a}{x - 1}$

and n is a non-negative integer

Ans: Basis step.

$$P(0): a = \frac{ax^{0+1} - a}{x - 1} = \frac{ax - a}{x - 1} = \frac{a(x - 1)}{x - 1} = a$$

$P(0)$ is true

Inductive step

Assume $p(k)$ is true, then

$$a + ax + ax^2 + \dots + ax^k = \frac{ax^{k+1} - a}{x-1}$$

$$\begin{aligned} \text{Then, } p(k+1) : a + ax + \dots + ax^k + ax^{k+1} &= \frac{ax^{k+1} - a}{x-1} + ax^{k+1} \\ &= \frac{ax^{k+1} - a + ax^{k+1}(x-1)}{x-1} \\ &= \frac{ax^{k+2} - a}{x-1}, \text{ which is true} \end{aligned}$$

$\therefore p(k+1)$ is true

$\therefore$ by induction $a + ax + ax^2 + \dots + ax^n = \frac{ax^{n+1} - a}{x-1}$

$$2, \quad 2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^n = \frac{1 - (-7)^{n+1}}{4}$$

n is a non-negative no.

Ans: Basic step

$$\text{let } p(n) : 2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^n = \frac{1 - (-7)^{n+1}}{4}$$

$$\text{Then } p(0) : 2 = \frac{(1) - (-7)^{0+1}}{4} = \frac{1+7}{4} = 2$$

$p(0)$ is true

Inductive step

Assume $p(k)$ is true

$$\text{then, } 2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^k = \frac{1 - (-7)^{k+1}}{4}$$

$$\begin{aligned}
 \text{Then } P(k+1) &: 2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2 \cdot (-7)^k + 2 \cdot (-7)^{k+1} \\
 &= \frac{1 - (-7)^{k+1}}{4} + 2 \cdot (-7)^{k+1} \\
 &= \frac{1 - (-7)^{k+1} + 8 \cdot (-7)^{k+1}}{4} \\
 &= \frac{1 + 7(-7)^{k+1}}{4}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1 - (-7)(-7)^{k+1}}{4} \\
 &= \frac{1 - (-7)^{k+2}}{4}, \text{ which is true}
 \end{aligned}$$

$\therefore \underline{P(k+1)}$ is true

$\therefore$ by induction, $\underline{2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2 \cdot (-7)^n = \frac{1 - (-7)^{n+1}}{4}}$

Q. Prove that $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \dots + n \cdot n! = (n+1)! - 1$, where n is a ≥ 1

$$\text{Let } P(n) : 1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \dots + n \cdot n! = (n+1)! - 1$$

Basis step

$$P(1) : 1 \cdot 1! = (1+1)! - 1 = 2! - 1 = 2 - 1 = 1 = (1+1)! - 1$$

$P(1)$ is true

Inductive step

Assume $P(k)$ is true.

$$\text{then, } 1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! = (k+1)! - 1$$

$$\text{then, } P(k+1) : 1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! + (k+1)! \cdot (k+1)$$

$$= ((k+1)! - 1) + (k+1)! \cdot (k+1)$$

$$= (k+1)! \cdot (k+1+1) - 1$$

$$= (k+2) \cdot (k+1)! - 1$$

$$= (k+2)! - 1, \text{ which is true}$$

$\therefore p(k+1)$ is true.

By induction, $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$

Q. Conjecture a formula for the sum of the first n positive odd integers. Then prove by using induction.

Ans: $1, 3, 5, 7, \dots, (2n-1)$

from A.P.

$$a_n = a + (n-1)d$$

$$= 1 + 2(n-1)$$

Sum,

$$1 = 1 = 1^2$$

$$1+3 = 4 = 2^2$$

$$1+3+5 = 9 = 3^2$$

$$1+3+5+7 = 16 = 4^2$$

$$\therefore 1+3+5+\dots+(2n-1) = n^2$$

$$\text{Let } p(n): 1+3+5+\dots+(2n-1) = n^2$$

Basis step.

$$p(1): (2n-1) = 2-1 = 1 = 1^2$$

$p(1)$ is true

Inductive step

Assume $p(k)$ is true,

$$\text{then, } p(k): 1+3+\dots+(2k-1) = k^2$$

$$\text{then, } p(k+1): 1+3+\dots+(2k-1) + (2(k+1)-1)$$

$$= k^2 + (2k+1)$$

$$= (k+1)^2, \text{ which is true}$$

$\therefore$ by induction, $1 + 3 + 5 + \dots + (2n-1) = n^2$

Q. Find a formula for $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n \cdot (n+1)}$

and prove by induction.

$$\frac{1}{1 \cdot 2} = \frac{1}{2}$$

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} = \frac{1}{2} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} = \frac{2}{3} + \frac{1}{12} = \frac{8+1}{12} = \frac{9}{12} = \frac{3}{4}$$

then, $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n \cdot (n+1)} = \frac{n}{n+1}$

let $p(n) : \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n \cdot (n+1)} = \frac{n}{n+1}$

$p(1) : \frac{1}{1 \cdot 2} = \frac{1}{2}$ which is true

$p(1)$ is true

Assume $p(k)$ is true, then

$$\left. \begin{aligned} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} &= \frac{k}{k+1} \end{aligned} \right\} \text{ is true}$$

then, $p(k+1) : \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$

$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{k^2 + 2k + 1}{(k+1)(k+2)}$$

$$= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}, \text{ which is true}$$

$\therefore p(k+1)$ is true

$\therefore$ by induction, $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

- Q. Use mathematical induction to ST $n < 2^n, \forall n \in \mathbb{Z}^+$
- Q. Use induction to PT $2^n < n!, \forall n \in \mathbb{Z}, n \geq 4$

Q. Use mathematical induction to ST $n < 2^n, \forall n \in \mathbb{Z}^+$

$$P(n) : n < 2^n, \forall n \in \mathbb{Z}^+$$

$P(1) : 1 < 2$, which is true.

$P(1)$ is true.

Assume $P(k)$ is true, then,

$$P(k) : k < 2^k$$

then, $P(k+1) : k+1 < 2^{k+1}$, WRT,

$$\frac{k+1}{2} = \frac{k}{2} \cdot 2$$

$$k < 2^k$$

$$\textcircled{x} 2 \Rightarrow 2k < 2^k \cdot 2$$

$$k+1 < 2^k + 1 \quad \text{--- (1)}$$

$$2^k \geq 2, \forall k \geq 1,$$

Since $2^k \geq 2$, $2^k \geq 1$ ($2^k \geq 2 \geq 1$)

$$1 \leq 2^k$$

$$2^k + 1 \leq 2^k + 2^k = 2^{k+1}$$

$$2^{k+1} \leq 2^{k+1} \text{ --- (2)}$$

From (1) & (2), ~~2^k~~ $k+1 < 2^{k+1}$,

$P(k+1)$ is true

then by induction, $n < 2^n \forall n \in \mathbb{Z}^+, n \geq 1$

Q. use induction to PT $2^n < n!$, $\forall n \in \mathbb{Z}, n \geq 4$

let $P(n) : 2^n < n! \forall n \in \mathbb{Z}, n \geq 4$

$$P(4) : 2^4 < 4!$$

$$\Rightarrow 16 < 24 \text{ which is true}$$

$P(4)$ is true

Assume $P(k)$ is true, then

$$2^k < k!$$

then, $P(k+1)$:

$$\text{WKT } 2^k < k!$$

$$\times 2, \quad 2^{k+1} < 2 \times k! \text{ --- (1)}$$

Since $k \geq 4$

$$k+1 \geq 5, \quad k+1 > 2 \quad (5 > 2)$$

$$2 < k+1$$

$$2 \times k! < (k+1)k! \text{ --- (2)}$$

$$= (k+1)!$$

$2^{k+1} < (k+1)!$, which is true.

$\therefore$ by induction, $2^n < n!$, $\forall n \in \mathbb{Z}, n \geq 4$

11/04/25
Q.

Hilbert's Grand Hotel: $\aleph_1$ countably infinite no. of rooms.

$$1 \cdot \aleph_1 \cdot \aleph_1 = \aleph_1$$

$$\aleph_1 \cdot \aleph_1 = \aleph_1 \quad 1 = 1$$

$$\aleph_1 \cdot \aleph_1 = \aleph_1 \cdot \aleph_1 = \aleph_1 \quad 2 = 2$$

$$\aleph_1 \cdot \aleph_1 = \aleph_1 \cdot \aleph_1 = \aleph_1 \quad 3 = 3$$

$$\aleph_1 \cdot \aleph_1 = \aleph_1 \quad \therefore$$

11/2.

Ex 4.

Ex. 8

consistent
a set where
all are true

(9) (10) (11) 12 21 23

8/04/25

RECURRENCE RELATION

The recurrence relation for a sequence is an equation for the relation that expresses a general term of the sequence in terms of one or more of the previous terms.

Eg:- 5, 10, 20, 40, ...

Here, $\frac{a_1}{a_0} = 2$, $\frac{a_2}{a_1} = 2$

$$\begin{matrix} a_0 & a_1 & a_2 & a_3 & \dots \\ \hline 5 & 10 & 20 & 40 & \dots \end{matrix}$$

$$\therefore \frac{a_{n+1}}{a_n} = 2$$

$$\Rightarrow a_{n+1} = 2a_n$$

$$\Rightarrow a_{n+1} - 2a_n = 0$$

This is the recurrence relation

(order is 1)
(largest suff. - smallest suff.)

Q. Solve the recurrence relation.

$$a_{n+1} = 3a_n, \quad n \geq 0, \quad a_0 = 5.$$

$$a_{n+1} = 3a_n$$

$$\Rightarrow n = 0, \quad a_1 = 3a_0$$

$$n = 1, \quad a_2 = 3a_1 = 3^2 a_0$$

$$n = 2, \quad a_3 = 3a_2 = 3^3 a_0$$

$\therefore a_n = 3^n a_0$, then general solution is $a_n = 5 \cdot 3^n$

2. $a_n = 7 \cdot a_{n-1}$, $n \geq 1$, $a_2 = 98$.

$$a_n = 7 \cdot a_{n-1}$$

$$n=1, a_1 = 7 \cdot a_0$$

$$n=2, a_2 = 7 \cdot a_1 = 7^2 \cdot a_0$$

$$n=3, a_3 = 7 \cdot a_2 = 7^3 \cdot a_0$$

$$\therefore \underline{a_n = 7^n \cdot a_0}$$

Given, $a_2 = 98 = 7^2 \cdot a_0$

$$\Rightarrow a_0 = \frac{98}{7^2} = \frac{98}{49} = 2$$

Then, general solution is $\underline{a_n = 2 \cdot 7^n}$

3. Solve $6a_n - 7a_{n-1} = 0$, $n \geq 1$, and $a_3 = 343$.

$$6a_n - 7a_{n-1} = 0 \Rightarrow a_n = \frac{7a_{n-1}}{6}$$

$$n=1, a_1 = \frac{7 \cdot a_0}{6}$$

$$n=2, a_2 = \frac{7 \cdot a_1}{6} = \left(\frac{7}{6}\right)^2 \cdot a_0$$

$$n=3, a_3 = \frac{7 \cdot a_2}{6} = \left(\frac{7}{6}\right)^3 \cdot a_0$$

then, $a_n = \left(\frac{7}{6}\right)^n \cdot a_0$

Given, $a_3 = 343 = \left(\frac{7}{6}\right)^3 \cdot a_0$

$$a_0 = \frac{343 \times 6^3}{7^3} = 6^3 = 216$$

then, general solution is, $\underline{a_n = 216 \cdot \left(\frac{7}{6}\right)^n}$

4. $3a_{n+1} - 4a_n = 0, n \geq 0$ and $a_1 = 5$

$$3a_{n+1} - 4a_n = 0$$

$$\Rightarrow a_{n+1} = \frac{4}{3}a_n$$

$$n=0, a_1 = \frac{4}{3}a_0$$

$$n=1, a_2 = \frac{4}{3}a_1 = \left(\frac{4}{3}\right)^2 a_0$$

$$n=2, a_3 = \frac{4}{3}a_2 = \left(\frac{4}{3}\right)^3 a_0$$

$$n=3, a_4 = \frac{4}{3}a_3 = \left(\frac{4}{3}\right)^4 a_0$$

$$\text{Then, } a_n = \left(\frac{4}{3}\right)^{n+1} a_0$$

~~The general solution is, $a_n = 5 \cdot \left(\frac{4}{3}\right)^{n-1}$~~

$$a_1 = \frac{4}{3}a_0 = 5$$

$$a_0 = \frac{15}{4}$$

The general solution is, $a_n = \frac{15}{4} \cdot \left(\frac{4}{3}\right)^{n+1}$

SECOND ORDER LINEAR HOMOGENEOUS RECURRENCE RELATION WITH CONSTANT COEFFICIENT

$$\rightarrow c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} = 0$$

let $a_n = c x^n$ be the solution.

Substituting in the eqn.

$$c_0 \cdot c x^n + c_1 \cdot c x^{n-1} + c_2 \cdot c x^{n-2} = 0$$

$$c_0 \cdot c x^n + c_1 \cdot c x^{n-1} + c_2 \cdot c x^{n-2} = 0$$

$$c_0 + \frac{c_1}{x} + \frac{c_2}{x^2} = 0$$

Take LCM x^2 , $\frac{c_0 x^2}{x^2} + \frac{x c_1}{x^2} + \frac{c_2}{x^2} = 0$

$\Rightarrow c_0 x^2 + c_1 x + c_2 = 0$, this is a quadratic in x .

on solving, we get 2 values for x , say x_1 and x_2 .

case 1: The roots are distinct & real.

Then, solution is

$$a_n = c_1 (x_1)^n + c_2 (x_2)^n$$

Eg:- roots are 4 and 5,

$$a_n = c_1 (4)^n + c_2 (5)^n$$

case 2: The roots are equal

$$\text{i.e., } x_1 = x_2 = x$$

then solution is

$$a_n = (c_1 + c_2 n) \cdot x^n$$

eg:- The roots are 4, 4,
 solution, $a_n = (c_1 + c_2 n) \cdot 4^n$

Q1. Solve the recurrence relation, $a_n = a_{n-1} + 2a_{n-2}$,
 $a_0 = 2, a_1 = 7$.

$$a_n - a_{n-1} - 2a_{n-2} = 0$$

Let $a_n = c \cdot x^n$ be the solution,

substituting,

$$c \cdot x^n - c \cdot x^{n-1} - 2c \cdot x^{n-2} = 0$$

$$c x^n - \frac{c x^n}{x} - 2 \frac{c x^n}{x^2} = 0$$

$$1 - \frac{1}{x} - \frac{2}{x^2} = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$x_1 = 2, x_2 = -1$$

$\therefore$ the solution is $a_n = c_1 (2)^n + c_2 (-1)^n$

Given, $a_0 = 2$

$$\Rightarrow a_0 = c_1 (2)^0 + c_2 (-1)^0$$

$$\boxed{2 = c_1 + c_2} \quad \text{--- (1)}$$

Given, $a_1 = 7$

$$\Rightarrow a_1 = c_1 (2)^1 + c_2 (-1)^1$$

$$\text{--- (2)} \quad \boxed{2c_1 - c_2 = 7} \quad \text{--- (2)}$$

$$\text{--- (1)} + \text{--- (2)} \Rightarrow \begin{pmatrix} 3c_1 = 9 \\ c_1 = 3 \end{pmatrix} \quad \leftarrow \left(\frac{2c_1 - c_2}{3} \right) \times \text{--- (1)}$$

$$\text{From --- (1), } \underline{\underline{c_2 = -1}}$$

Then, solution is, $a_n = \underline{\underline{3 \cdot (2)^n - 1 \cdot (-1)^n}}$

Q2. Solve the recurrence relation with generate the numbers 0, 1, 1, 2, 3, 5, 8, 13, ...

Here, the recurrence relation ~~are~~ is

$$a_n = a_{n-1} + a_{n-2}$$

$$a_n - a_{n-1} - a_{n-2} = 0$$

let the solution be $a_n = c \cdot x^n$,
substituting,

$$c \cdot x^n - c \cdot \frac{x^n}{x} - c \cdot \frac{x^n}{x^2} = 0$$

$$1 - \frac{1}{x} - \frac{1}{x^2} = 0$$

$$x^2 - x - 1 = 0$$

$$(-1)^2 - 4(1)$$

$$= 1 - 4 = -3$$

we get the solution,

$$x_1 = \frac{1+\sqrt{5}}{2}, \quad x_2 = \frac{1-\sqrt{5}}{2}$$

$$\frac{1 \pm \sqrt{5}}{2}$$

The solution is, $a_n = c_1 \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \cdot \left(\frac{1-\sqrt{5}}{2}\right)^n$

$$a_0 = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^0 + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^0$$

$$\boxed{c_1 + c_2 = 0} \text{ --- ①}$$

$$a_1 = \boxed{c_1 \left(\frac{1+\sqrt{5}}{2}\right) + c_2 \left(\frac{1-\sqrt{5}}{2}\right) = 1} \text{ --- ②}$$

$$\textcircled{1} \times \left(\frac{1+\sqrt{5}}{2}\right) \Rightarrow \left(\frac{1+\sqrt{5}}{2}\right) c_1 + \left(\frac{1+\sqrt{5}}{2}\right) c_2 = 0 \textcircled{1}$$

$$c_2 \left(\frac{1-\sqrt{5}}{2} - \frac{1+\sqrt{5}}{2}\right) = 1$$

$$c_2 \left(\frac{-\sqrt{5} \times 2}{2}\right) = 1$$

$$\underline{\underline{c_2 = -\frac{1}{\sqrt{5}}}}$$

$$\text{then, } \underline{\underline{c_1 = \frac{1}{\sqrt{5}}}}$$

Then solution, $a_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n \rightarrow$ this is the explicit formula

Q3. $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$, $a_0 = 2$, $a_1 = 5$, $a_2 = 15$.

$$a_n - 6a_{n-1} + 11a_{n-2} - 6a_{n-3} = 0$$

substituting, $a_n = c \cdot x^n$

$$\Rightarrow c \cdot x^n - 6c \cdot \frac{x^n}{x} + 11 \cdot c \cdot \frac{x^n}{x^2} - 6 \cdot c \cdot \frac{x^n}{x^3} = 0$$

$$1 - \frac{6}{x} + \frac{11}{x^2} - \frac{6}{x^3} = 0$$

$$x^3 - 6x^2 + 11x - 6 = 0$$

solution, $x_1 = 1$, $x_2 = 2$, $x_3 = 3$.

then solution is, $a_n = c_1(1)^n + c_2(2)^n + c_3(3)^n$

$n=0$, $a_0 = c_1 + c_2 + c_3 = 2$ — ①, $c_1 = 1$

$a_1 = c_1 + 2c_2 + 3c_3 = 5$ — ②, $c_2 = -1$

$a_2 = c_1 + 4c_2 + 9c_3 = 15$ — ③, $c_3 = 2$.

① $\rightarrow$ ②

then solution is, $a_n = (1)^n - (2)^n + 2 \cdot (3)^n$

Q4. $a_n = 6a_{n-1} - 9a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 6$

$$a_n - 6a_{n-1} + 9a_{n-2} = 0$$

let the soln. be $a_n = c \cdot x^n$

substituting,

$$c \cdot x^n - 6c \cdot \frac{x^n}{x} + 9c \cdot \frac{x^n}{x^2} = 0$$

$$1 - \frac{6}{x} + \frac{9}{x^2} = 0$$

$$\Rightarrow x^2 - 6x + 9 = 0$$

Here, roots are $x_1 = x_2 = x = 3$

$$\therefore a_n = (C_1 + C_2 n) \cdot x^n$$

$$a_n = (C_1 + C_2 n) \cdot 3^n$$

$$\text{Given, } a_0 = 1 = (C_1 + C_2) \cdot 3^0$$

$$C_1 + C_2 = 1$$

$$a_1 = 6$$

$$\Rightarrow (C_1 + C_2) \cdot 3 = 6$$

$$C_1 + C_2 = 2$$

$$\Rightarrow \underline{C_1 = C_2 = 1}$$

$\therefore$ solution is $a_n = (1+n) \cdot 3^n$

Q5. Solve $a_n = 6a_{n-1} - 11a_{n-2} + 3a_{n-3} - 3a_{n-2} - a_{n-3}$

$$a_0 = 1, a_1 = 2, a_2 = -1$$

Ans

$$a_n + 3a_{n-1} + 3a_{n-2} + a_{n-3} = 0$$

Let $a_n = c \cdot x^n$ is the solution,

substituting,

$$c \cdot x^n + 3 \cdot c \cdot \frac{x^n}{x} + 3 \cdot c \cdot \frac{x^n}{x^2} + c \cdot \frac{x^n}{x^3} = 0$$

$$1 + \frac{3}{x} + \frac{3}{x^2} + \frac{1}{x^3} = 0$$

$$x^3 + 3x^2 + 3x + 1 = 0$$

$$x_1 = x_2 = x_3 = -1$$

Solution is, $a_n = (C_1 + C_2 n + C_3 n^2) \cdot (-1)^n$

$$a_0 = 1 \neq 0 = (c_1 + c_2^0 + c_3^0) \cdot (-1)^0$$

$$\underline{c_1 = 1}$$

$$a_1 = -2 = (c_1 + c_2 + c_3) \cdot (-1)$$

$$(1 + c_2 + c_3) = 2$$

$$c_2 + c_3 = 1 \quad \text{--- (1)}$$

$$a_2 = -1 = (c_1 + 2c_2 + 4c_3) \cdot (1)$$

$$1 + 2c_2 + 4c_3 = -1$$

$$2c_2 + 4c_3 = -2$$

$$2(c_2 + 2c_3) = -2$$

$$c_2 + 2c_3 = -1 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \Rightarrow -c_3 = 2$$

$$\underline{\underline{c_3 = -2}}$$

$$c_2 - 4 = -1$$

$$\underline{\underline{c_2 = 3}}$$

Then solution is, $\underline{\underline{a_n = (1 + 3n - 2n^2) \cdot (-1)^n}}$

09/04/25

NON-HOMOGENEOUS

$f(n)$

$a_n^{(p)}$

$$a_n^{(p)} = a_n^{(h)} + a_n^{(p)}$$

①

$k\delta^n$

$A\delta^n$

$$\text{or } f(n) = \delta^n = (1)^n$$

②

n

$A_1 n + A_0$

③

n^2

$A_1 n^2 + A_2 n + A_3$

④

k

A

$$\Sigma = (\Sigma \cdot A) \otimes I + (I \otimes \Sigma \cdot A) \otimes I + \Sigma \cdot A$$

$$I = A \cdot \Sigma + \Sigma \cdot A \cdot \Sigma + \Sigma \cdot A$$

$$I = A \Sigma + A \Sigma + A \Sigma$$

$$I = A \Sigma$$

$$\Sigma I = A$$

Q1. Solve the recurrence relation $a_{n+2} + 3a_{n+1} + 2a_n = 3^n$,
 $n \geq 0$, $a_0 = 0$, $a_1 = 1$,

Ans: This is non-homogeneous.

$$a_n = a_n^{(h)} + a_n^{(p)}$$

First, we find $a_n^{(h)}$,

$$a_{n+2} + 3a_{n+1} + 2a_n = 0 \quad (h)$$

~~$a_{n+2} = -3a_{n+1} - 2a_n$~~ , let $a_n = c \cdot x^n$ be the solution.

$$c x^{n+2} + 3c x^{n+1} + 2c x^n = 0$$

$$x^2 + 3x + 2 = 0$$

$$(x+2)(x+1) = 0$$

$$x = -2, -1, \text{ different solution.} \leftarrow (2) - (1)$$

$$\therefore a_n^{(h)} = c_1 (-2)^n + c_2 (-1)^n$$

Given, $a_0 = 0$

$$= c_1 (-2)^0 + c_2 (-1)^0$$

$$\Rightarrow c_1 + c_2 = 0$$

To find $a_n^{(p)}$,

$$\text{Here, } f(n) = 3^n$$

$$\therefore a_n^{(p)} = A \cdot 3^n$$

Substituting,

$$a_{n+2} + 3a_{n+1} + 2a_n = 3^n$$

$$A \cdot 3^{n+2} + 3(A \cdot 3^{n+1}) + 2(A \cdot 3^n) = 3^n$$

$$A \cdot 3^2 + 3 \cdot A \cdot 3 + 2 \cdot A = 1$$

$$9A + 9A + 2A = 1$$

$$20A = 1$$

$$A = 1/20$$

$$\therefore a_n^{(p)} = \frac{1}{20} \cdot (3^n)$$

$\therefore$ general solution is, $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = c_1 (-2)^n + c_2 (-1)^n + \frac{1}{20} (3^n)$$

Given, $a_0 = 0$ and $a_1 = 1$

$$a_0 = c_1 (-2)^0 + c_2 (-1)^0 + \frac{1}{20} (3^0)$$

$$a_0 = c_1 + c_2 + \frac{1}{20} = 0$$

$$c_1 + c_2 = -\frac{1}{20} \quad \text{--- (1)}$$

$$a_1 = c_1 (-2)^1 + c_2 (-1)^1 + \frac{1}{20} (3^1)$$

$$= -2c_1 - c_2 + \frac{3}{20} = 1$$

$$-2c_1 - c_2 = 1 - \frac{3}{20}$$

$$-2c_1 - c_2 = \frac{17}{20} \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow -c_1 = \frac{16}{20} \Rightarrow -\frac{4}{5}$$

$$c_1 = -\frac{4}{5}$$

$$c_2 = -\frac{1}{20} + \frac{4}{5} = \frac{15-1}{20} = \frac{14}{20} = \frac{7}{10}$$

$$c_1 = -\frac{4}{5}, \quad c_2 = \frac{7}{10}$$

$$\text{Then, } a_n = -\frac{4}{5} (-2)^n + \frac{7}{10} (-1)^n + \frac{1}{20} (3^n)$$

Q2. Solve $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$

Q3. Solve $a_{n+2} - 8a_{n+1} + 16a_n = 8 \cdot (5^n)$, $n \geq 0$, $a_0 = 12$, $a_1 = 5$.

2. This is non-homogeneous.

$$a_n = a_n^{(h)} + a_n^{(p)}$$

First, we find $a_n^{(h)}$

$$a_n - 5a_{n-1} + 6a_{n-2} = 0$$

Let $a_n = c \cdot \delta^n$ be the solution,

$$c\delta^n - 5c\delta^{n-1} + 6c\delta^{n-2} = 0$$

$$1 - \frac{5}{\delta} + \frac{6}{\delta^2} = 0$$

$$x^2 - 5x + 6 = 0$$

$$(x-3)(x-2) = 0$$

$$x = 3, 2$$

$$\text{Then, } a_n = c_1(3)^n + c_2(2)^n$$

To find a_n ,

$$\text{Here, } f(n) = 7^n$$

$$\therefore a_n = A \cdot 7^n$$

$$a_n - 5a_{n-1} + 6a_{n-2} = 7^n$$

$$\Rightarrow A \cdot 7^n - 5(A \cdot 7^{n-1}) + 6(A \cdot 7^{n-2}) = 7^n$$

$$\Rightarrow A \cdot 7^n - 5 \cdot A \cdot \frac{7^n}{7} + 6 \cdot A \cdot \frac{7^n}{7^2} = 7^n$$

$$\Rightarrow A - \frac{5A}{7} + \frac{6A}{49} = 1$$

$$\Rightarrow 49A - 35A + 6A = 49$$

$$20A = 49$$

$$A = \frac{49}{20}$$

$$a_n = \frac{49}{20} \cdot (7^n)$$

$\therefore$ General solution is $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = c_1(3)^n + c_2(2)^n + \frac{49}{20} \cdot 7^n$$

3. $a_{n+2} - 8a_{n+1} + 16a_n = 8 \cdot (5)^n$
 This is non-homogeneous.

First, we find $a_n^{(h)}$, $\mathcal{L} = \mathcal{D}$, $0 < n$, $\mathcal{N} \mathcal{B} = 1 - 4\mathcal{D} \mathcal{S} = \mathcal{N} \mathcal{D}$ A.D

$$a_{n+2} - 8a_{n+1} + 16a_n = 0$$

let $a_n = c\delta^n$ be the solution, $\mathcal{N} \mathcal{B} = 1 - 4\mathcal{D} \mathcal{S} = \mathcal{N} \mathcal{D}$ A.D

$$c\delta^n \cdot \delta^2 - 8 \cdot c\delta^n \cdot \delta + 16 \cdot c\delta^n = 0$$

$$\delta^2 - 8\delta + 16 = 0$$

$$(\delta - 4)(\delta - 4) = 0$$

$$\delta_1 = \delta_2 = \delta = 4$$

$$\therefore a_n^{(h)} = \underline{(c_1 + c_2 n) \cdot 4^n}$$

To find, $a_n^{(p)}$

$$f(n) = 8 \cdot (5)^n$$

$$\therefore a_n^{(p)} = A \cdot 5^n$$

substituting in $a_{n+2} - 8a_{n+1} + 16a_n = 8 \cdot (5)^n$

$$A \cdot 5^n \cdot 25 - 8 \cdot A \cdot 5^n \cdot 5 + 16 \cdot A \cdot 5^n = 8(5^n)$$

$$25A - 40A + 16A = 8 \quad \mathcal{N} \mathcal{B} = (\mathcal{N}) \mathcal{E}$$

$$\underline{A = 8}$$

$$\underline{A = 8}$$

$$\therefore a_n^{(p)} = 8 \cdot 5^n$$

Then, general solution, $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = (c_1 + c_2 n) \cdot 4^n + 8 \cdot 5^n$$

Given, $a_0 = 12$, $a_1 = 5$

$$a_0 = c_1 \cdot 4^0 + 8(5^0)$$

$$= c_1 + 8 = 12$$

$$c_1 = 4$$

$$a_1 = (c_1 + c_2) \cdot 4 + 8 \cdot 5 = 5$$

$$(4 + c_2) \cdot 4 = 5 - 40 = -35$$

$$4 + c_2 = \frac{-35}{4}$$

$$c_2 = \frac{-35 - 16}{4} = \frac{-51}{4}$$

41
40

56
25
31

$$\therefore \text{general solution, } a_n = \left(4 - \frac{5}{4}n\right) \cdot 4^n + 8 \cdot 5^n$$

Q.4. $a_n = 3a_{n-1} + 2n$, $n \geq 0$, $a_1 = 3$.

$$a_n - 3a_{n-1} = 2n$$

This is non-homogeneous.

To find, $a_n^{(h)}$,

$$a_n - 3a_{n-1} = 0$$

let $a_n^{(h)} = c \cdot x^n$ be the solution,

$$c \cdot x^n - 3c \cdot \frac{x^n}{x} = 0$$

$$1 - \frac{3}{x} = 0$$

$$x - 3 = 0$$

$$\underline{x = 3.}$$

Then, $\underline{a_n^{(h)} = c \cdot 3^n}$

To find $a_n^{(p)}$

$$f(n) = 2n$$

$$\therefore a_n^{(p)} = A_1 n + A_0,$$

Substituting in $a_n - 3a_{n-1} = 2n$,

$$(A_1 n + A_0) - 3(A_1 (n-1) + A_0) = 2n$$

$$A_1 n + A_0 - 3A_1 (n-1) - 3A_0 = 2n$$

$$A_1 (n - 3n + 3) + 2A_0 = 2n$$

$$-2A_1 n + 3A_1 + 2A_0 = 2n \Rightarrow 3A_1 - 2nA_1 - 2A_0 = 2n$$

Equating the coefficient of 'n' on both sides,

$$-2A_1 = 2$$

$$\underline{A_1 = -1}$$

Equating the constant term,

$$3A_1 - 2A_0 = 0$$

$$-3 - 2A_0 = 0$$

$$-2A_0 = 3$$

$$A_0 = -3/2$$

$$\therefore a_n^{(p)} = -n - \frac{3}{2}$$

$$\therefore \text{general solution, } a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_n = C \cdot 3^n - n - \frac{3}{2}$$

Given, $a_1 = 3$

$$a_1 = C \cdot 3 - 1 - \frac{3}{2} = 3$$

$$3C = 3 + 1 + \frac{3}{2} = \frac{11}{2}$$

$$3C = \frac{6 + 2 + 3}{2} = \frac{11}{2}$$

$$3C = \frac{11}{2} \Rightarrow C = \frac{11}{6}$$

Then, general solution, $a_n = \frac{11}{6} \cdot 3^n - n - \frac{3}{2}$

Q5. $a_{n+2} + 4a_{n+1} + 4a_n = 7$, $a_0 = 1$, $a_1 = 2$.

To find a_n ,

$$a_{n+2} + 4a_{n+1} + 4a_n = 0$$

let $a_n = c \cdot x^n$ be solution,

$$c \cdot x^{n+2} + 4c \cdot x^{n+1} + 4c \cdot x^n = 0$$

$$x^2 + 4x + 4 = 0$$

$$(x+2)(x+2) = 0$$

$$x_1 = x_2 = x = -2$$

Then $a_n^{(h)} = (c_1 + c_2 n) \cdot (-2)^n$

To find $a_n^{(p)}$,

$$f(n) = 7$$

$$\therefore a_n^{(p)} = A.$$

Substituting in $a_{n+2} + 4a_{n+1} + 4a_n = 7$

$$A + 4A + 4A = 7$$

$$9A = 7$$

$$A = 7/9$$

$$\therefore a_n^{(p)} = 7/9$$

$\therefore$ general solution, $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = (c_1 + c_2 n) \cdot (-2)^n + \frac{7}{9}$$

Given, $a_0 = 1, a_1 = 2$.

$$a_0 = c_1 (-2)^0 + \frac{7}{9} = 1$$

$$c_1 = 1 - \frac{7}{9} = \frac{2}{9}$$

$$c_1 = 2/9$$

$$a_1 = (c_1 + c_2) (-2) + \frac{7}{9} = 2$$

$$\left(\frac{2}{9} + c_2\right) (-2) = 2 - \frac{7}{9} = \frac{18-7}{9} = \frac{11}{9}$$

$$\frac{2}{9} + c_2 = \frac{11}{18}$$

$$c_2 = \frac{11}{18} - \frac{2}{9} = \frac{11-4}{18} = \frac{7}{18}$$

∴ general solution is,

$$a_n = \left(\frac{2}{9} - \frac{5}{6}n \right) \cdot (-2)^n + \frac{7}{9}$$

Q6. $a_{n+1} - 2a_n = 2^n$, $n \geq 0$, $a_0 = 1$

To find a_n ,

$$a_{n+1} - 2a_n = 0$$

Let $a_n = c \cdot x^n$ be the solution,

$$c \cdot x^n \cdot x - 2 \cdot c x^n = 0$$

$$x - 2 = 0$$

$$x = 2$$

Then, $a_n^{(h)} = c \cdot (2)^n$

To find $a_n^{(p)}$,

$$f(n) = 2^n$$

$$\therefore a_n^{(p)} = A n (2^n)$$

Substituting in $a_{n+1} - 2a_n = 2^n$,

$$A \cdot (2^n) \cdot (2) - 2(A \cdot 2^n) = 2^n \quad A(n+1)(2^n)(2) - 2(A n 2^n) = 2^n$$

$$2A - 2A = 1 \quad 2A(n+1) - 2A n = 1$$

$$2A n + 2A - 2A n = 1$$

$$A = 1/2$$

$$a_n^{(p)} = \frac{1}{2} n (2^n)$$

∴ general solution, $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = c \cdot 2^n + \frac{1}{2} n 2^n$$

Given, $a_0 = 1$

$$a_0 = c = 1$$

∴ general solution, $a_n = 2^n + \frac{n}{2} (2^n)$

if x and value in $f(n)$ (here 2^n) are same, different process.

then $a_n^{(p)} = A n^2 (2^n)$

• if $x = 2$, 2 , then $a_n^{(p)} = A n^2 (2^n)$

∴ cases.

Q7 $a_{n+2} - 4a_{n+1} + 3a_n = -200, n \geq 0, a_0 = 3000, a_1 = 2300$

To find $a_n^{(h)}$

$$a_{n+2} - 4a_{n+1} + 3a_n = 0$$

Let $a_n = c \cdot x^n$ be the solution,

$$c x^n \cdot x^2 - 4 c x^n \cdot x + 3 c x^n = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x_1 = 3, x_2 = 1$$

$$a_n^{(h)} = c_1(3)^n + c_2(1)^n$$

To find $a_n^{(p)}$

$$f(n) = -200$$

~~Wrong~~

$$\therefore a_n = A$$

Substituting in $a_{n+2} - 4a_{n+1} + 3a_n = -200$

$$\Rightarrow -A - 4A + 3A = -200$$

becoz, $a_{n+2} - 4a_{n+1} + 3a_n = -200(1)^n$ — roots are 1 and 1

$$\text{so, } a_n^{(p)} \neq A$$

$$\text{instead, } a_n^{(p)} = An$$

$$a_n^{(p)} = An$$

Substituting in $a_{n+2} - 4a_{n+1} + 3a_n = -200$,

$$A(n+2) - 4A(n+1) + 3(An) = -200$$

$$An + 2A - 4An - 4A + 3An = -200$$

$$-2A = -200$$

$$A = 100$$

$$\therefore a_n^{(p)} = 100n$$

General solution, $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = c_1(3)^n + c_2(1)^n + 100n$$

Given, $a_0 = 3000, a_1 = 3300,$

$$a_0 = c_1 + c_2 = 3000 \quad \text{--- (1)}$$

$$a_1 = 3c_1 + c_2 + 100 = 3300 \quad \text{--- (2)}$$

$$3c_1 + c_2 = 3200$$

$$2c_1 + 3000 = 3200$$

$$2c_1 = 200$$

$$\underline{c_1 = 100}, \quad \underline{c_2 = 2900}$$

General solution, $a_n = \underline{100(3^n)} + \underline{2900(1)^n} + 100n$

$$a_n = \underline{100(3^n)} + \underline{2900} + 100n$$

Q8. $a_n + 4a_{n-1} + 4a_{n-2} = 5(-2)^n, n \geq 0, a_0 = 1, a_1 = 0.$

Q9. $a_{n+1} - a_n = 3n^2 - n, n \geq 0, a_0 = 3$

Q10. $a_n = 4a_{n-1} - 4a_{n-2}, n \geq 0, a_0 = 1$

Q11. $a_{n+1} - a_n = 2n + 3, n \geq 0, a_0 = 1$

Q12. $a_{n+1} = a_n + n, n \geq 0, a_2 = 1$

Q13. $a_n = a_{n-1} + a_{n-2} + 1, n \geq 0, a_0 = 0, a_1 = 0.$

Q14. $a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3} + n4^n, a_0 = 2, a_1 = 0.$

$$Z = [(A + nA^2 - 8n) + (1 + n8 - 8n)8 - 8n] A \leftarrow$$

$$Z = [A + nA^2 - 8n + 8 - 8nA + 8n - 8n] A$$

$$Z = A8$$

$$Z/8 = A$$

Q8. $a_n + 4a_{n-1} + 4a_{n-2} = 5(-2)^n, n \geq 0, a_0 = 1, a_1 = 0$

To find a_n ,

$$a_n + 4a_{n-1} + 4a_{n-2} = 0$$

Let $a_n = c \cdot x^n$ be the solution,

$$cx^n + 4 \frac{cx^n}{x} + 4 \frac{cx^n}{x^2} = 0$$

$$x^2 + 4x + 4 = 0$$

$$(x+2)^2 = 0$$

$$x_1 = x_2 = x = -2.$$

$$\therefore a_n = (c_1 + c_2 n)(-2)^n$$

To find a_n ,

$$f(n) = 5(-2)^n$$

$$\therefore a_n = An^2(-2)^n$$

Substituting $a_n + 4a_{n-1} + 4a_{n-2} = 5(-2)^n$

$$\Rightarrow An^2(-2)^n + 4(A(n-1)^2(-2)^{n-1}) + 4A(n-2)^2(-2)^{n-2} = 5(-2)^n$$

$$\Rightarrow An^2(-2)^n + 4A \frac{(-2)^n}{(-2)} (n-1)^2 + 4A \frac{(-2)^n}{(-2)^2} (n-2)^2 = 5(-2)^n$$

$$\Rightarrow An^2 + \frac{4A(n-1)^2}{(-2)} + \frac{4A(n-2)^2}{(-2)^2} = 5$$

$$\Rightarrow A [n^2 - 2(n^2 - 2n + 1) + (n^2 - 4n + 4)] = 5$$

$$A [n^2 - 2n^2 + 4n - 2 + n^2 - 4n + 4] = 5$$

$$2A = 5$$

$$A = 5/2$$

$$\therefore a_n^{(p)} = \frac{5}{2} n^2 (-2)^n$$

the general solution is $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = (C_1 + C_2 n) (-2)^n + \frac{5}{2} n^2 (-2)^n$$

Given, $a_0 = 1$ and $a_1 = 0$.

$$a_0 = C_1 (-2)^0 + 0 = 1$$

$$C_1 = 1$$

$$a_1 = (C_1 + C_2) (-2) + \frac{5}{2} (-2) = 0$$

$$(1 + C_2) + \frac{5}{2} = 0$$

$$C_2 = -\frac{5}{2} - 1 = -\frac{5+2}{2} = -\frac{7}{2}$$

$$\therefore a_n = (1 - \frac{7}{2}n) (-2)^n + \frac{5}{2} n^2 (-2)^n$$

Q1. $a_{n+1} - a_n = 3n^2 - n, n \geq 0, a_0 = 3.$

To find a_n ,

$$a_{n+1} - a_n = 0$$

let $a_n^{(h)} = c \cdot \delta^n$, be the solution.

$$c\delta^{n+1} - c\delta^n = 0$$

$$\delta - 1 = 0$$

$$\delta = 1$$

$$\therefore a_n^{(h)} = c \cdot (1)^n$$

To find $a_n^{(p)}$,

$$f(n) = 3n^2 - n$$

Since, $\delta = 1$,

$$a_n^{(p)} = (A_1 n^2 + A_2 n + A_3) n$$

Substituting in $a_{n+1} - a_n = 2n^2 - n$,

$$(A_1(n+1)^2 + A_2(n+1) + A_3)(n+1) - (A_1n^2 + A_2n + A_3)n = 2n^2 - n$$

$$A_1(n^3 + 1^3 + 3n^2 + 3) + A_2(n^2 + 2n + 1) + A_3(n+1) - (A_1n^3 + A_2n^2 + A_3n) = 2n^2 - n$$

equating n^2 , $3A_1 + A_2 - A_2 = 3$

equating n , $2A_2 + A_3 - A_3 = -1$

equating constants, $A_1 + 3A_1 + A_2 + A_3 = 0$

$$4A_1 + A_2 + A_3 = 0$$

$$4 \cdot \frac{1}{2} + A_2 + A_3 = 0$$

$$2 + A_2 + A_3 = 0$$

$$A_2 + A_3 = -2$$

$$A_2 = -2 - A_3$$

$\therefore a_n^{(p)} = (n^2 - 2n + 1)n$

$\therefore$ General solution, $a_n = a_n^{(h)} + a_n^{(p)}$

$$= c \cdot (1)^n + (n^2 - 2n + 1)n$$

Given, $a_0 = 3$

$a_0 = c = 3$

$\therefore$ general solution, $a_n = 3(1)^n + n^3 - 2n^2 + n$

$$a_n = 3 + n(n-1)^2$$

210 (3) $a_n = 4a_{n-1} - 4a_{n-2} + (n+1)2^n$
 to find $a_n^{(h)}$,

$$a_n - 4a_{n-1} + 4a_{n-2} = 0$$

let $a_n^{(h)} = c\delta^n$ be the solution,

$$c\delta^n - 4c\frac{\delta^n}{\delta} + 4c\frac{\delta^n}{\delta^2} = 0$$

$$\delta^2 - 4\delta + 4 = 0$$

$$\underline{\underline{\delta_1 = \delta_2 = 2}}$$

$$\therefore a_n^{(h)} = \underline{(c_1 + c_2 n)(2)^n}$$

to find $a_n^{(p)}$,

$$f(n) = (n+1)2^n$$

$$\therefore a_n^{(p)} = n^2(A_1 n + A_0) \cdot 2^n$$

Substituting in $a_n - 4a_{n-1} + 4a_{n-2} = (n+1)2^n$,

$$n^2(A_1 n + A_0)2^n - 4[(n-1)^2(A_1(n-1) + A_0)2^{n-1}] + 4[(n-2)^2(A_1(n-2) + A_0)2^{n-2}] = (n+1)2^n$$

$$n^2(A_1 n + A_0) - 2[(n-1)^2(A_1(n-1) + A_0)] + [(n-2)^2(A_1(n-2) + A_0)] = (n+1)$$

$$A_1 n^3 + A_0 n^2 - 2[A_1(n-1)^3 + A_0(n-1)^2] + [A_1(n-2)^3 + A_0(n-2)^2] = (n+1)$$

$$A_1 n^3 + A_0 n^2 - 2[A_1(n^3 - 3n^2 + 3n - 1) + A_0(n^2 - 2n + 1)] + [A_1(n^3 - 6n^2 + 12n - 8) + A_0(n^2 - 4n + 4)] = n+1$$

comparing n^3 , $A_1 - 2A_1 + A_1 = 0$

$$\text{comparing } n^2, \quad A_0 + 6A_1 - 2A_0 - 6A_1 + A_0 = 0 \quad (N)$$

$$\text{comparing } n, \quad -6A_1 + 4A_0 + 12A_1 - 4A_0 = 1 \quad (N)$$

$$A_1 = 1/6 \quad (N)$$

$$\text{comparing const, } 2A_1 - 2A_0 - 8A_1 + 4A_0 = 0 \quad (N)$$

$$-6A_1 + 2A_0 = 1$$

$$-1 + 2A_0 = 1$$

$$A_0 = 1 \quad (N)$$

$$\therefore a_n^{(p)} = n^2 \left(\frac{1}{6}n + 1 \right) \cdot 2^n \quad (N)$$

~~Given~~

General solution,

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_n = (C_1 + C_2 n) \cdot (2^n) + \left(\frac{1}{6}n^3 + n^2 \right) \cdot 2^n$$

$$[(1+N)S - (0A + (1-N)A)] S = [(0A + (1-N)A)] S$$

$$(1+N)S = [(0A + (1-N)A)] S$$

$$[(0A + (1-N)A)] S = (0A + (1-N)A) S$$

$$(1+N)S = [(0A + (1-N)A)] S$$

$$(1+N)S$$

$$[(1+N)0A + (1-N)A] S = 0A + (1-N)A$$

$$(1+N)S = [(1+N)0A + (1-N)A] S$$

$$[(1+N)0A + (1-N)A] S = 0A + (1-N)A$$

$$(1+N)S = [(1+N)0A + (1-N)A] S$$

$$S = 1A + 1AS - 1A \quad (N)$$

④ $a_{n+1} - a_n = 2n + 3, n \geq 0, a_0 = 1$

To find $a_n^{(h)}$,

$$a_{n+1} - a_n = 0$$

let $a_n^{(h)} = c \cdot x^n$ be the solution.

$$c x^{n+1} - c x^n = 0 \quad (x^{n+1} - x^n) = 0$$

$$x - 1 = 0$$

$$\underline{x = 1}$$

$$\therefore a_n^{(h)} = c(1)^n$$

To find $a_n^{(p)}$,

$$f(n) = 2n + 3$$

Since, $x = 1$,

$$\therefore a_n^{(p)} = (A_1 n + A_0)n$$

Substituting in $a_{n+1} - a_n = 2n + 3$,

$$(A_1(n+1) + A_0)(n+1) - (A_1 n + A_0)n = 2n + 3$$

$$A_1(n^2 + 2n + 1) + A_0(n+1) - A_1 n^2 - A_0 n = 2n + 3$$

Comparing,

$$A_1 - A_1 = 0$$

$$A_1 + A_0 = 3$$

$$\cancel{2A_1} + A_0 = 2$$

$$\underline{A_0 = 2}$$

$$2A_1 + A_0 = 2$$

$$2A_1 + \cancel{A_0} = 2$$

$$\underline{A_1 = 1}$$

$$\underline{a_n^{(p)} = (n+2)n}$$

General solution, $a_n = a_n^{(h)} + a_n^{(p)} = c(1)^n + (n+2)n$

Given, $a_0 = 1$

$$\Rightarrow c = 1$$

General solution, $a_n = 1 + n(n+2)$
 $= n^2 + 2n + 1$

$$a_n = (n+1)^2$$

Abhinand

SEMINAR

①

→ You can access the internet from the campus if you are a CS major or you're not a freshman

p: you can access the internet from campus

q: you are a CS major

r: you are a freshman

$$p \rightarrow (q \vee \neg r)$$

Abhinand

② You cannot ride the rollercoaster if you are under 4 ft tall unless you are older than 16.

p: you can ride the rollercoaster

q: you are under 4ft tall.

r: You are older than 16.

If you are under 4ft tall and not 16 yrs old then you cannot ride the rollercoaster

$$(q \wedge \neg r) \rightarrow \neg p$$

③ the automated reply cannot be sent when the file system is full.

p: The automated reply can be sent

q: The file system is full.

$$q \rightarrow \neg p$$

④ "The Diagnostic message is stored in the buffer or it is retransmitted."

"The diagnostic message is not stored in the buffer".
 "the diagnostic message is stored in the buffer, then it is ^{re}transmitted."

p: The diagnostic msg is stored

q: It is retransmitted.

$(p \vee q) \wedge (\neg p \vee q) \wedge \neg p$ is true, then p - false

$p \vee q$ is true

$\neg p \rightarrow q$ $\Rightarrow$ false $\vee$ q is true

$\Rightarrow$ then q is true
 false $\rightarrow$ true is true.

hence, system is consistent.

⑤ The diagnostic message is not transmitted.

$$p \vee q$$

$$\neg p$$

$$p \rightarrow q$$

$$\neg q$$

⑥ Boolean search — search result

AND - narrow range

OR - widen range

NOT - exclude a term

" "

- exact phrase result

() - convert the logic of the search

(cat AND dog) OR food

cat food and dog food.

Sidhanth

⑦

empty empty treasure
in trunk 2

P_i : trunk i has treasure

$$[(\neg P_1) \wedge (\neg P_2) \wedge (\neg P_2)] \vee [(\neg P_1) \wedge (\neg P_2) \wedge (\neg P_2)] \\ \vee [(\neg P_1) \wedge (\neg P_2) \wedge P_2]$$

$$(\neg P_1 \wedge P_2 \wedge \neg P_2) \vee (P_1 \wedge \neg P_2) \vee (P_1 \wedge P_2 \wedge P_2)$$

$$(\neg P_1 \wedge F_0) \vee (P_1 \wedge \neg P_2) \vee (P_1 \wedge P_2)$$

$$P_1 \wedge (\neg P_2 \vee P_2)$$

$$P_1 \wedge T$$

P_1 , treasure is in trunk 1

dhruv

⑧ An island that has two kinds of inhabitants, knights - who always tell the truth, and their opp. knaves - who always lie. You encounter two people, A and B. What are A and B if

A says B is a knight

B says the two of us are opp-types.

p : A is a knight
 q : B is knight
 $\neg p$: A is knave
 $\neg q$: B is knave.

A says B is a knight.

$$p \leftrightarrow q$$

B says the two us are opposite types.

If B is knight, $p \oplus q$ is true.

If B is knave, $p \oplus q$ is false.

$$q \leftrightarrow p \oplus q$$

$$\neg p \leftrightarrow p \oplus q$$

both are knaves.

$$\alpha \quad \alpha \quad \alpha \quad \alpha \quad \alpha \quad \alpha \quad \alpha \quad \alpha \quad \alpha \quad \alpha$$

HW. 12. ⑤ $a_{n+1} = a_n + n, n \geq 0, a_2 = 1$

To find $a_n^{(h)}$,

$$a_{n+1} - a_n = 0$$

let $a_n^{(h)} = c \cdot x^n$ be the solution,

$$c x^{n+1} - c x^n = 0, 0 \leq n, 1 + x - x^2 = 0$$

$$x - 1 = 0$$

$$x = 1$$

$$\therefore a_n^{(h)} = c \cdot (1)^n$$

To find $a_n^{(p)}$,

$$f(n) = n$$

$$\therefore a_n^{(p)} = (A_1 n + A_0) n$$

Substituting in $a_{n+1} - a_n = n$

$$\Rightarrow [A_1(n+1) + A_0](n+1) - (A_1 n + A_0)n = n$$

$$A_1(n^2 + 2n + 1) + A_0(n+1) - A_1 n^2 - A_0 n = n$$

$$\text{comparing, } 2A_1 + A_0 - A_0 = 1$$

$$A_1 = 1/2.$$

$$A_1 + A_0 = 0$$

$$A_0 = -1/2$$

$$\therefore a_n^{(p)} = \left(\frac{1}{2}n - \frac{1}{2}\right)n$$

$$\therefore \text{general solution, } a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_n = c \cdot (1)^n + \frac{1}{2}n(n-1)$$

$$\text{Given, } a_2 = 1$$

$$a_2 = c(1)^2 + \frac{1}{2} \times 2(2-1)$$

$$= c + 1 = 1$$

$$\underline{c = 0}$$

$$\therefore \text{general solution, } a_n = \underline{\frac{1}{2}n(n-1)}$$

Q.13

$$(6) a_n = a_{n-1} + a_{n-2} + 1, n \geq 0, a_0 = 0, a_1 = 0$$

To find $a_n^{(h)}$,

$$a_n - a_{n-1} - a_{n-2} = 0$$

$$\text{let } a_n^{(h)} = c \cdot x^n \text{ be the solution, } (1) \cdot c = \frac{(n)}{n}$$

$$c x^n - \frac{c x^n}{x} - \frac{c x^n}{x^2} = 0$$

$$x^2 - x - 1 = 0$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

$$\therefore a_n^{(h)} = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

To find $a_n^{(p)}$,

$$f(n) = 1$$

$$\therefore a_n^{(p)} = A$$

substituting in $a_n - a_{n-1} - a_{n-2} = 1$

$$A - A - A = 1 \Rightarrow A = -1$$

$$\underline{A = -1}$$

$$\therefore \underline{a_n^{(p)} = -1}$$

$\therefore$ general solution, $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = c_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + c_2 \left(\frac{1-\sqrt{5}}{2} \right)^n - 1$$

Given, $a_0 = a_1 = 0$

$$a_0 = c_1 + c_2 - 1 = 0$$

$$c_1 + c_2 = 1 \quad \text{--- (1)}$$

$$c_1 - c_2 = 1/\sqrt{5} \quad \text{--- (2)}$$

$$2c_1 = \frac{\sqrt{5}+1}{\sqrt{5}}$$

$$\underline{c_1 = \frac{\sqrt{5}+1}{2\sqrt{5}}}$$

$$c_2 = 1 - \frac{\sqrt{5}+1}{2\sqrt{5}} = \frac{2\sqrt{5}-\sqrt{5}-1}{2\sqrt{5}}$$

$$\underline{c_2 = \frac{\sqrt{5}-1}{2\sqrt{5}}}$$

$$\therefore \text{general solution, } a_n = \left(\frac{\sqrt{5}+1}{2\sqrt{5}} \right) \left(\frac{1+\sqrt{5}}{2} \right)^n + \left(\frac{\sqrt{5}-1}{2\sqrt{5}} \right) \left(\frac{1-\sqrt{5}}{2} \right)^n - 1$$

Q.14

(7)

$$a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3} + n4^n, \quad a_0 = 2, \quad a_1 = 0$$

To find $a_n^{(h)}$

$$a_n - 7a_{n-1} + 16a_{n-2} - 12a_{n-3} = 0$$

let $a_n^{(h)} = c \cdot x^n$ be the solution.

$$c \cdot x^n - 7 \cdot \frac{c x^n}{x} + 16 \cdot \frac{c x^n}{x^2} - 12 \cdot \frac{c x^n}{x^2} = 0$$

$$x^3 - 7x^2 + 16x - 12 = 0$$

$$x_1 = 3, \quad x_2 = x_3 = 2.$$

$$\therefore a_n^{(h)} = (C_1 + C_2 n)(2)^n + C_3(3)^n$$

To find $a_n^{(p)}$,

$$f(n) = n4^n$$

$$\therefore a_n^{(p)} = (A_1 n + A_0) \cdot 4^n$$

$$\text{Substituting in } a_n - 7a_{n-1} + 16a_{n-2} - 12a_{n-3} = n4^n$$

$$(A_1 n + A_0) \cdot 4^n - 7[A_1(n-1) + A_0] \cdot 4^{n-1} + 16[A_1(n-2) + A_0] \cdot 4^{n-2} - 12[A_1(n-3) + A_0] \cdot 4^{n-3} = n4^n$$

$$A_1 n + A_0 - \frac{7}{4}(A_1(n-1) + A_0) + \frac{16}{16}[A_1(n-2) + A_0]$$

$$- \frac{12}{43}[A_1(n-3) + A_0] = n$$

$$A_1 n + A_0 - \frac{7}{4}(A_1 n - A_1 + A_0) + A_1 n - 2A_1 + A_0$$

$$- \frac{3}{16}[A_1 n - 3A_1 + A_0] = n$$

comparing,

$$A_1 - \frac{7}{4}A_1 + A_1 - \frac{3}{16}A_1 = 1$$

$$\frac{-28 - 3A_1}{16} = 1$$

$$- \frac{31}{16} A_1 = 1$$

$$A_1 = -\frac{16}{31}$$

$$A_0 + \frac{7}{4}A_1 - \frac{7}{4}A_0 - 2A_1 + A_0 + \frac{9}{16}A_1 - \frac{3}{16}A_0 = 0$$

$$16A_0 + 28A_1 - 28A_0 - 32A_1 + 16A_0 + 9A_1 - 3A_0 = 0$$

$$32A_0 - 28A_0 - 3A_0 + 28A_1 - 32A_1 - 9A_1 = 0$$

$$-9A_1 = 0$$

$$A_0 - 7A_1 = 0$$

$$7A_1 = A_0$$

$$A_0 = 7 \times \frac{16}{31}$$

15. $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$, where $F(n) = 3^n$

(i) $= 2^n$ (vi) $= (n^2+1)2^n$
(ii) $= 2^n$ (vii) $= (n+1)2^n$
(iii) $= n3^n$ (viii) $= n \cdot 2^n$
(iv) $= n2^n$ (ix) $= k$
(v) $= n^2 2^n$

⊕ To find, $a_n^{(h)}$,

$$a_n - 6a_{n-1} + 9a_{n-2} = 0$$

let $a_n^{(h)} = c \cdot \delta^n$ be the solution,

$$c\delta^n - 6c \cdot \frac{\delta^n}{\delta} + 9c \cdot \frac{\delta^n}{\delta^2} = 0$$

$$\delta^2 - 6\delta + 9 = 0$$

$$\delta_1 = \delta_2 = 3$$

$$\therefore a_n^{(h)} = \underline{(c_1 + c_2 n) \cdot 3^n}$$

① To find $a_n^{(p)}$,

$$F(n) = 3^n$$

$$\therefore a_n^{(p)} = A n^2 3^n$$

② $\therefore a_n^{(p)} = A \cdot 2^n$

③ $\therefore a_n^{(p)} = (A_1 n + A_0) n^2 \cdot 3^n$

④ $\therefore a_n^{(p)} = (A_1 n + A_0) 2^n$

⑤ $a_n^{(p)} = (A_1 n^2 + A_2 n + A_3) 2^n$

⑥ $a_n^{(p)} = (A_1 n^2 + A_2 n + A_3) n^2 \cdot 3^n$

⑦ $a_n^{(p)} = (A_1 n + A_0) 2^n$

⑧ $a_n^{(p)} = (A_1 n + A_0) 2^n$

⑨ $a_n^{(p)} = A$

Generating Functions

$$(a+b)^n = a^n + n c_1 a^{n-1} b + n c_2 a^{n-2} b^2 + n c_3 a^{n-3} b^3 + \dots + b^n$$

$$(1+x)^n = 1 + nC_1 x + nC_2 x^2 + nC_3 x^3 + \dots + x^n$$

$$(1-x)^n = 1 - nC_1 x + nC_2 x^2 - nC_3 x^3 + \dots + (-x)^n$$

no. of terms = power + 1

EXTENSION OF BINOMIAL TH.

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1 \cdot 2} x^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} x^3 + \dots$$

$$(1+x)^{-1} = 1 - x + \frac{(-1)(-2)}{1 \cdot 2} x^2 + \frac{(-1)(-2)(-3)}{1 \cdot 2 \cdot 3} x^3 + \dots$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$$

$$(1+x)^{-2} = 1 - 2x + \frac{(-2)(-3)}{1 \cdot 2} x^2 + \frac{(-2)(-3)(-4)}{1 \cdot 2 \cdot 3} x^3 + \dots$$

$$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

$$(1-x)^{-1} = 1 - 1(-x) + \frac{(-1)(-2)}{1 \cdot 2} (-x)^2 + \dots$$

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

GENERATING FUNCTIONS

Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers

then $G(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$

is called generating function.

$$\text{i.e., } G(x) = \sum_{k=0}^{\infty} a_k x^k$$

STANDARD GENERATING FUNCTIONS

1. We know, $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$

$$\frac{1}{(1-x)} = 1 + x + x^2 + x^3 + \dots$$

$\frac{1}{1-x}$ is the generating function of the sequence
1, 1, 1, ...

2. $\frac{d}{dx}(1-x)^{-1} = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots$

$\frac{1}{(1-x)^2}$ is the generating function of the sequence
1, 2, 3, 4, ...

3. Multiplying by x ,
 $\frac{x}{(1-x)^2} = 0 + x + 2x^2 + 3x^3 + 4x^4 + \dots$

$\frac{x}{(1-x)^2}$ is the generating function of the sequence
0, 1, 2, 3, 4, ...

4. $\frac{d}{dx} \frac{x}{(1-x)^2} = \frac{(1-x^2) \times 1 - x \times 2(1-x)(-1)}{(1-x)^4}$

$$= \frac{(1-x)^2 + 2x(1-x)}{(1-x)^4} = \frac{(1-x)[(1-x) + 2x]}{(1-x)^4}$$

$$= \frac{1+x}{(1-x)^3}$$

$$\frac{1+x}{(1-x)^3} = 1 + 4x + 9x^2 + 16x^3 + \dots$$

$\frac{1+x}{(1-x)^3}$ is the generating function of the sequence
 $1^2, 2^2, 3^2, 4^2, \dots$

5. multiplying by α .

$$\frac{\alpha(1+\alpha)}{(1-\alpha)^3} = 0 + \alpha + 4\alpha^2 + 9\alpha^3 + 16\alpha^4 + \dots$$

$\frac{\alpha(1+\alpha)}{(1-\alpha)^3}$ is the generating function of the sequence $0^2, 1^2, 2^2, 3^2, 4^2, \dots$

6. we know, $\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \alpha^3 + \dots$

Here, we put, $\alpha = 2y$,

$$\frac{1}{1-2y} = 1 + 2y + 4y^2 + 8y^3 + \dots$$

Sequence is $2^0, 2^1, 2^2, 2^3, \dots$

$\frac{1}{1-2y}$ is the generating function of the sequence $2^0, 2^1, 2^2, 2^3, \dots$

7. Put $\alpha = 3y$, $\frac{1}{1-3y} = 1 + 3y + 9y^2 + 27y^3 + \dots$

Sequence is $3^0, 3^1, 3^2, 3^3, \dots$

$\frac{1}{1-3y}$ is the generating function of the sequence $3^0, 3^1, 3^2, 3^3, \dots$

8. we know, $\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \alpha^3 + \dots$

Multiplying by 2, $\frac{2}{1-\alpha} = 2 + 2\alpha + 2\alpha^2 + 2\alpha^3 + \dots$

$\frac{2}{1-\alpha}$ is the generating function of the sequence $2, 2, 2, 2, \dots$

9. multiplying by 3,

$$\frac{3}{1-\alpha} = 3 + 3\alpha + 3\alpha^2 + 3\alpha^3 + \dots$$

10. We know $\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \alpha^3 + \dots$

$\frac{1}{1+\alpha}$ is the generating function of the sequence
1, -1, 1, -1, ...

11. We know $\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \alpha^3 + \dots$

$\frac{1}{1+\alpha} = 1 - \alpha + \alpha^2 - \alpha^3 + \dots$

Adding, $\frac{1}{1-\alpha} + \frac{1}{1+\alpha} = \frac{1+\alpha+1-\alpha}{1-\alpha^2}$

$= \frac{2}{1-\alpha^2}$

$\frac{2}{1-\alpha^2} = 2 + 2\alpha^2 + 2\alpha^4 + \dots$

$\frac{1}{1-\alpha^2} = 1 + 0\alpha + \alpha^2 + 0\alpha^3 + \alpha^4 + \dots$

$\frac{1}{1-\alpha^2}$ is the generating function of the sequence
1, 0, 1, 0, 1, ...

12. $\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \alpha^3 + \dots$

$\frac{1}{1+\alpha} = 1 - \alpha + \alpha^2 - \alpha^3 + \dots$

Subtracting, $\frac{1+\alpha - 1+\alpha}{1-\alpha^2} = \frac{2\alpha}{1-\alpha^2}$

$\frac{2\alpha}{1-\alpha^2} = 2\alpha + 2\alpha^3 + 2\alpha^5 + \dots$

$\frac{\alpha}{1-\alpha^2} = \alpha + \alpha^3 + \alpha^5 + \dots$

$$\frac{\alpha}{1-\alpha^2}$$

$0, 1; 0, 1, 0, \dots$

Q. solve $a_k = 3a_{k-1}$, $k \geq 1$, $a_0 = 2$

$$a_k - 3a_{k-1} = 0.$$

let $a_k = c \cdot \delta^k$

$$c\delta^k - 3c\frac{\delta^k}{\delta} = 0$$

$$8 - 3 = 0$$

$$\delta = 3.$$

$$\therefore a_k = c \cdot (3)^k$$

$$a_0 = 2 = c(3)^0$$

$$\underline{\underline{a_k = 2 \cdot (3)^k}}$$

using generating fn. method,

$$a_k = 3 a_{k-1}$$

$$\sum_{k=1}^{\infty} a_k x^k = 3 \sum_{k=1}^{\infty} a_{k-1} x^k$$

$$G(x) - a_0 = \sum_n G(n)$$

Given, $a_0 = 2$,

$$G(x) [1 - 3x] = 2$$

$$G(x) = \frac{2}{1-3x}$$

$$= 2(1-3\alpha)^{-1} = 2(1-3\alpha)^{-1} = 2(1-3\alpha)^{-1} = 2 \quad \text{--- (2)}$$

$$= 2(1 + 3x + 3^2x^2 + \dots)$$

$$\sum_{k=0}^{\infty} a_k x^k = x \sum_{k=0}^{\infty} 3^k x^k$$

$$\therefore \underline{\underline{a_k = 2.3^k}}$$

Q. Solve the recurrence relation,
by generative fn.

$$a_k = 7 \cdot a_{k-1}, \quad a_0 = 5$$

$$\sum_{k=1}^{\infty} a_k x^k = 7 \sum_{k=1}^{\infty} a_{k-1} x^k$$

$$G(x) - a_0 = 7x G(x)$$

$$G(x) - 7x G(x) = a_0$$

$$G(x) (1 - 7x) = 5$$

$$G(x) = \frac{5}{1-7x}$$

$$= 5(1-7x)^{-1}$$

$$= 5(1 + 7x + 7^2 x^2 + \dots)$$

$$\sum_{k=0}^{\infty} a_k x^k = 5 \sum_{k=0}^{\infty} 7^k x^k$$

$$\therefore a_k = 5 \cdot 7^k$$

11/04/25

① $a_n = 3a_{n-1}, k \geq 1, a_0 = 2$

② $a_k = 7a_{k-1}, a_0 = 5$

③ $a_n = 8a_{n-1} + 10^{n-1}, a_1 = 9$

④ $a_k = 3a_{k-1} + 4^{k-1}, a_0 = 1$

⑤ $a_k = 3a_{k-1} + 2, a_0 = 1$

⑥ $a_k = 5a_{k-1} - 6a_{k-2}, a_0 = 6, a_1 = 30$

Ans:

③ $a_n = 8a_{n-1} + 10^{n-1}, a_1 = 9$

$$\sum_{n=1}^{\infty} a_n x^n = 8 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} 10^{n-1} x^n$$

$$a_1 = 8a_0 + 10^0 = 9$$

$$8a_0 = 8$$

$$a_0 = 1$$

$$x + 10x^2 + 100x^3 + \dots$$

$$a_0 x$$

$$a_1 x + a_2 x^2 + \dots$$

$$G(x) - a_0 = 8x G(x) + \sum_{n=1}^{\infty} 10^{n-1} x^n$$

$$G(x) - 1 = 8x G(x) + \sum_{n=1}^{\infty} 10^{n-1} x^n$$

$$\begin{aligned} G(x) (1-8x) &= \sum_{n=1}^{\infty} 10^{n-1} x^n + 1 \\ &= 1 + x + 10x^2 + 100x^3 + \dots \\ &= 1 + x (1 + 10x + 10^2 x^2 + \dots) \end{aligned}$$

$$= 1 + x [1 - 10x]^{-1}$$

$$G(x) (1-8x) = \frac{1-10x + 10x}{1-10x} = \frac{1-9x}{1-10x}$$

$$G(x) = \frac{1-9x}{(1-10x)(1-8x)}$$

(using partial fraction)

Now, $\frac{1-9x}{(1-10x)(1-8x)} = \frac{A}{(1-10x)} + \frac{B}{(1-8x)}$

Multiplying by,

$$(1-9x) = A(1-8x) + B(1-10x)$$

comparing, $-9 = -8A - 10B$ and $1 = A + B$ (318 410)

$$8A + 10B = 9$$

$$8A + 8B = 8$$

$$2B = 1$$

$$B = 1/2 \text{ and } A = 1/2$$

$$\therefore \frac{1-9x}{(1-10x)(1-8x)} = \frac{1/2}{1-10x} + \frac{1/2}{1-8x}$$

Then, $G(x) = \frac{1}{2} \left[\frac{1}{1-10x} + \frac{1}{1-8x} \right]$

$$= \frac{1}{2} \left[(1-10x)^{-1} + (1-8x)^{-1} \right]$$

$$= \frac{1}{2} \left[(1 + 10x + 10^2 x^2 + \dots) + (1 + 8x + 8^2 x^2 + \dots) \right]$$

$$\sum_{n=0}^{\infty} a_n x^n = \frac{1}{2} \sum_{n=0}^{\infty} 10^n x^n + \frac{1}{2} \sum_{n=0}^{\infty} 8^n x^n \quad (x) \text{D}$$

$$\therefore a_n = \frac{1}{2} (10^n + 8^n) \quad (x) \text{D}$$

$$\textcircled{4} \quad a_k = 3a_{k-1} + 4^{k-1}, \quad a_0 = 1$$

$$\sum_{k=1}^{\infty} a_k x^k = \sum_{k=1}^{\infty} 3a_{k-1} x^k + \sum_{k=1}^{\infty} 4^{k-1} x^k$$

$$G(x) - a_0 = 3x G(x) + [x + 4x^2 + \dots]$$

$$G(x)(1-3x) = 1 + x + 4x^2 + \dots$$

$$= 1 + x [1 + 4x + 4^2 x^2 + 4^3 x^3 + \dots]$$

$$\frac{1}{(x) \text{D}} = \frac{1 + x [1 - 4x]^{-1} - 1}{(x) \text{D} (x) \text{D} - 1}$$

$$= \frac{1 - 4x + x}{1 - 4x}$$

$$(x) \text{D} - 1 \equiv \frac{1 - 3x}{1 - 4x} A = (x) \text{D} - 1$$

$$G(x) = \frac{1}{1 - 4x} = (1 - 4x)^{-1}$$

$$G(x) = 1 + 4x + 4^2 x^2 + \dots$$

$$\sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} 4^k x^k$$

$$\therefore a_k = 4^k$$

$$\left[\frac{1}{x^2 - 1} + \frac{1}{x^2 + 1} \right] \text{D} = (x) \text{D}$$

$$\left[\frac{1}{(x^2 - 1)} + \frac{1}{(x^2 + 1)} \right] \frac{1}{2}$$

$$\left[\frac{1}{(x^2 - 1)} + \frac{1}{(x^2 + 1)} \right] \frac{1}{2}$$

$$\textcircled{5} \quad a_k = 3a_{k-1} + 2, \quad a_0 = 1$$

$$\sum_{k=1}^{\infty} a_k x^k = 3 \sum_{k=1}^{\infty} a_{k-1} x^k + \sum_{k=1}^{\infty} 2x^k$$

$$G(x) - a_0 = 3x G(x) + [2x + 2x^2 + \dots]$$

$$G(x) [1 - 3x] = 1 + 2x + 2x^2 + \dots$$

$$(1 - 3x) G(x) = 1 + x(2 + 2x + 2x^2 + \dots)$$

$$= 1 + x \times \frac{2}{1-x}$$

$$= \frac{1 - x + 2x}{1 - x}$$

$$G(x) = \frac{1+x}{1-3x}$$

$$G(x) = \frac{1+x}{(1-3x)(1-x)}$$

$$\text{Now, } \frac{1+x}{(1-3x)(1-x)} = \frac{A}{1-3x} + \frac{B}{1-x}$$

$$(1+x) = A(1-x) + B(1-3x)$$

$$\text{comparing, } 1 = A + B \quad \text{and} \quad 1 = -A - 3B \quad \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow 2 = -2B$$

$$\underline{\underline{B = -1}} \quad \text{and} \quad \underline{\underline{A = 2}}$$

$$\frac{1+x}{(1+3x)(1-x)} = \frac{2}{1-3x} + \frac{-1}{1-x}$$

$$\therefore G(x) = \frac{2}{1-3x} + \frac{-1}{1-x}$$

$$= 2(1-3x)^{-1} - 1(1-x)^{-1} = 2(1+3x+3^2x^2+\dots) - 1(1+x+x^2+\dots)$$

$$\sum_{k=0}^{\infty} a_k x^k = 2 \sum_{k=0}^{\infty} 3^k x^k - 1 \sum_{k=0}^{\infty} 1^k x^k$$

$$\therefore a_k = \underline{\underline{2 \cdot 3^k - 1}}$$

6. $a_k = 5a_{k-1} - 6a_{k-2}$, $a_0 = 6$, $a_1 = 30$

$$\sum_{k=2}^{\infty} a_k x^k = 5 \sum_{k=2}^{\infty} a_{k-1} x^k - 6 \sum_{k=2}^{\infty} a_{k-2} x^k$$

(next page)

$$G(x) - a_0 - a_1 x = 5x G(x) - 6x^2 G(x)$$

$$(6x^2 - 5x + 1) G(x) = a_0 + a_1 x$$

$$G(x) = \frac{6 + 30x}{6x^2 - 5x + 1}$$

$$= \frac{6(1 - 5x)}{2x(3x - 1)(2x + 1)}$$

$$G(x) = \frac{6(1 - 5x)}{(3x - 1)(2x + 1)}$$

Now, $\frac{1 - 5x}{(3x - 1)(2x + 1)} = \frac{A}{(3x - 1)} + \frac{B}{(2x + 1)}$

$$1 - 5x = A(2x + 1) + B(3x - 1)$$

comparing,

$$1 = A + B \text{ and } -5 = 2A + 3B$$

$$A + B = 1$$

$$2(A + B) + B = -5$$

$$2 + B = -5$$

$$A = 1 + B$$

$$\underline{\underline{B = -3}}$$

$$\frac{1}{(x-1)} = \frac{1}{(x-1)} + \frac{0}{(x-1)} = \frac{1}{(x-1)}$$

$$\frac{1}{x-1} + \frac{0}{x^2-1} = \frac{1}{(x-1)}$$

$$(1 - x^2)(x-1) = (x-1)(1-x) + (x^2-1)(x-1)$$

$$\sum_{k=0}^{\infty} x^k = 1 + \sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} x^k$$

$$G(x) - a_0 - a_1 x = 5x(G(x) - a_0) - 6x^2 G(x)$$

$$G(x) - a_0 - a_1 x = 5xG(x) - 5xa_0 - 6x^2 G(x)$$

$$G(x)[6x^2 - 5x + 1] = a_0 + a_1 x - 5xa_0$$

$$= 6 + 30x - 30x$$

$$G(x) = \frac{6}{6x^2 - 5x + 1}$$

$$= \frac{6}{(3x-1)(2x-1)}$$

Now, $\frac{6}{(3x-1)(2x-1)} = \frac{A}{(3x-1)} + \frac{B}{(2x-1)}$

$$6 = A(2x-1) + B(3x-1)$$

comparing, $6 = -A - B$ and $0 = 2A + 3B$

$$A + B = -6 \quad 2(A+B) + B = 0$$

$$(2x-1)(-1) = -1$$

$$A = -18$$

$$-12 + B = 0 \quad B = 12$$

$$G(x) = \frac{-18}{(3x-1)} + \frac{12}{(2x-1)}$$

$$= \frac{18}{(1-3x)} - \frac{12}{(1-2x)}$$

$$= 18(1 + 3x + 3^2x^2 + \dots) - 12(1 + 2x + 2^2x^2 + \dots)$$

$$\sum_{k=0}^{\infty} a_k x^k = 18 \sum_{k=0}^{\infty} 3^k x^k - 12 \sum_{k=0}^{\infty} 2^k x^k$$

$$a_k = 18 \cdot 3^k - 12 \cdot 2^k$$

$$(1 + 3x + 3^2x^2 + \dots) = \frac{1}{1-3x}$$

$$= \frac{1}{1-3x} - \frac{1}{1-2x}$$

→ we know $a^2 - b^2 = (a-b)(a+b)$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^4 - b^4 = (a-b)(a^3 + a^2b + ab^2 + b^3)$$

$$\text{then, } a^{n+1} - b^{n+1} = (a-b)(a^n + a^{n-1}b + a^{n-2}b^2 + \dots + b^n)$$

Here, we put $a=1$ and $b=\alpha$

$$1^{n+1} - \alpha^{n+1} = (1-\alpha)(1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^n)$$

$$1 - \alpha^{n+1} = (1-\alpha)(1 + \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^n)$$

$$\boxed{\frac{1 - \alpha^{n+1}}{1 - \alpha} = 1 + \alpha + \alpha^2 + \dots + \alpha^n}$$

Here, the sequence is $1, 1, 1, \dots, 1, 0, 0, 0, \dots$
($n+1$ times)

(1 occurs $n+1$ times)

the generating function is $\frac{1 - \alpha^{n+1}}{1 - \alpha}$

Q. what is the generating fn. for the sequence

(i) $1, 1, 1, 1, 1, 1$

$\frac{1 - \alpha^6}{1 - \alpha}$ is the generating fn.

(2) $2, 2, 2, 2, 2, 2$

$$G(\alpha) = 2 + 2\alpha + 2\alpha^2 + 2\alpha^3 + 2\alpha^4 + 2\alpha^5$$

$$= 2(1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5)$$

$$= 2 \left(\frac{1 - \alpha^6}{1 - \alpha} \right) \text{ is the generating fn.}$$

(iii) 0, 2, 2, 2, 2, 2, 2, 0, 0, 0, 0, 0

$$G(x) = 0 + 2x + 2x^2 + 2x^3 + 2x^4 + 2x^5 + 2x^6$$

$$= 0 + 2x(1 + x + x^2 + x^3 + x^4 + x^5)$$

$$= 2x \left(\frac{1-x^6}{1-x} \right) \text{ is the generating fn.}$$

MODELLING

→ solution formula

Q. Find an explicit formula for the fibonacci numbers.

modelling
* staircase problem

- comp. interest
- rabbits & fibonacci no.s.
- tower of hanoi puzzle.
- counting bit strings.

① Existence of inverse: for each $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$

$$e = a \cdot a^{-1} = a^{-1} \cdot a$$

② Σ , the set of integers under addition is a group

① Closure property

$$a, b \in \Sigma \Rightarrow a+b \in \Sigma$$

∴ satisfies closure property

② Associative property

$$(a+b)+c = a+(b+c) \text{ for } a, b, c \in \Sigma$$

∴ satisfies

③ Identity element

$$a \in \Sigma \Rightarrow a+0 = 0+a = a$$

$$0 \in \Sigma$$

∴ 0 is the identity element.

STRONG INDUCTION

Q1. ST 'If n is an integer greater than 1, then n can be written as product of primes' by strong induction.

$P(n)$: n can be written as product of primes.

$P(2)$: 2 is a prime.

$P(3)$: 3 is a prime.

$P(4)$: $4 = 2 \times 2$, prime

$P(5)$: 5 is a prime

$P(6)$: $6 = 2 \times 3$

$\therefore$ It is true, $\forall j, 2 \leq j \leq k$

Now, we have to check $k+1$.

If $k+1$ is prime, $\therefore P(k+1)$ is true.

If $k+1$ is composite, it can be written as,

product of 2 +ve integers a and b

with $2 \leq a \leq b \leq (k+1)$

Since a and b are integers, b/w 2 and k , we can use the inductive hypothesis to write both a and b as the product of primes.

$\therefore \underline{P(k+1)}$ is true

